

Notes: Stein (JF, 2002)

Information Production and Capital Allocation: Decentralized versus Hierarchical Firms

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1 Big picture

- **Question.** How does organizational design affect the production of information and the allocation of capital across competing investment projects?
- **Setting.** The paper compares two organizational forms in a simple model with two operating units and externally constrained finance:
 - **Decentralization:** small stand-alone firms run by line managers who both gather information and control capital.
 - **Hierarchy:** a large integrated firm with division managers below a CEO who can shift funds across divisions.
- **Core trade-off.** Hierarchy has an advantage in reallocating funds across divisions, but it may also distort line-manager incentives to gather information in the first place.
- **Main conceptual contribution.** The paper shows that the answer depends crucially on whether project information is *soft* or *hard*.
- **What is soft information?** Information is soft when the line manager can use it personally but cannot credibly transmit it upward to a CEO or outward to outside financiers.
- **What is hard information?** Information is hard when at least some of it can be documented and passed upward inside the firm.
- **Main result under soft information.** Decentralization can beat hierarchy because line managers have stronger incentives to become informed when they know they will retain some capital to act on what they learn.
- **Main result under hard information.** Hierarchy dominates decentralization because line managers can use verifiable good news to argue for a larger internal budget, so separation of authority from expertise now strengthens rather than weakens incentives.
- **Additional implication 1.** Holding size and scope fixed, softer information tends to favor flatter organizations, because CEO effort and line-manager effort become strategic substitutes.
- **Additional implication 2.** If information hardness is endogenous, hierarchy can create excessive bureaucracy: managers may work too hard on documentation and report writing rather than on the socially most valuable soft-information production.
- **Empirical motivation.** The motivating application is bank consolidation. The paper argues that relationship-based small-business lending relies heavily on soft information, so large bank hierarchies may be comparatively bad at it.
- **Economic takeaway.** Large firms are not always worse, and small firms are not always better. The right organizational form depends on how information can be produced, transmitted, and acted upon.

2 Model environment and key objects

2.1 Organizational forms and actors

There are two operating units, i and j , each run by its own division manager. These units can either remain separate or be integrated.

- **Decentralization.** Each unit is a stand-alone firm. The line manager is effectively the CEO of his own small firm.
- **Hierarchy.** The two units are brought under one roof, and the division managers cede formal capital-allocation authority to a top-level CEO.

Interpretation. The whole paper is a comparison between these two governance structures. The key question is how information and authority should be matched.

2.2 Projects, states, and production technology

Each division contains two investment projects. Each project can receive either zero, one, or two units of capital, and each project can be in either a good state G or a bad state B . States are independent across all projects and occur with probability $1/2$.

A good project yields

$$g(1) > 0 \quad \text{with one unit of capital,} \quad g(2) > 0 \quad \text{with two units.} \quad (1)$$

A bad project yields

$$b(1), \quad b(2). \quad (2)$$

The paper imposes the restrictions

$$g(2) < 2g(1), \quad (3)$$

$$g(2) > g(1) + b(1), \quad (4)$$

$$b(1) = 0, \quad (5)$$

$$-1 < b(2) < 0, \quad (6)$$

$$g(2) + b(2) < 0. \quad (7)$$

Interpretation.

- $g(2) < 2g(1)$ means decreasing returns to scale in the good state.
- $g(2) > g(1) + b(1)$ means that if one project is good and the other bad, concentrating two units on the good project is better than splitting one unit each.
- $b(1) = 0$ is just a normalization.
- $g(2) + b(2) < 0$ ensures that too much uninformed capital can be value-destroying. This creates external financing constraints and makes internal capital allocation meaningful.

2.3 Research technology of division managers

If division manager i exerts effort e_i , he learns the states of both projects in his division with probability $p(e_i)$, where $p(\cdot)$ is increasing and concave and takes values in $[0, 1]$. Manager j has the same technology.

Interpretation. Information is not free. Research effort must be motivated. This is why organizational design matters at all.

2.4 Utility and empire-building preferences

Every agent k has utility

$$U_k = (Y_k + I_k) - \gamma e_k, \quad (8)$$

where Y_k is expected net output from the assets under the agent's control, I_k is the amount initially invested under that agent's authority, and γ is the cost of effort.

Interpretation. Agents care about the size of their capital budget as well as how profitably it is used. So they are empire builders, but conditional on having a given amount of capital they still want to allocate it efficiently.

2.5 External financiers and why credit is fixed

Outside investors cannot directly observe managers' signals. Since managers always prefer more capital to less and cannot be made fully responsive to monetary incentives, they cannot be induced to reveal their private information truthfully to outsiders.

Interpretation. External finance is therefore given in fixed, uncontingent amounts. This is what creates room for internal capital markets and for organizational form to matter.

2.6 Soft information versus hard information

The paper's central distinction is between two communication technologies.

- **Soft information.** A manager may genuinely learn something useful, but cannot credibly pass it upward. The information is real but nonverifiable.
- **Hard information.** The manager can document at least part of what he learns, so that the CEO can process and use it.

Interpretation. Under soft information, expertise is hard to separate from authority. Under hard information, expertise can travel upward and be combined with centralized authority.

2.7 Why the CEO may know more than outside investors

A hierarchy only becomes interesting if the CEO can sometimes do better than the outside market in deciding how much total capital to allocate and where to send it. Stein gives two rationales:

- outside investors may free ride in information acquisition;
- the CEO's stronger control rights make information more valuable to her, creating a complementarity between authority and research incentives.

Interpretation. This is a key modeling move. The CEO is not assumed to be omniscient. She simply has better incentives than outsiders to learn about divisional prospects.

3 Models and theoretical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper compares organizations along two margins:

- **Within-division allocation:** once a division gets a budget, does the informed line manager steer funds toward the better project?
- **Across-division allocation:** can the organization move capital from a weak division to a strong one?

The main hypotheses are:

- hierarchy helps on the across-division margin because the CEO can reallocate internally;
- decentralization may help on the within-division margin if line managers need strong incentives to gather soft information;
- if information is hard, hierarchy can improve both margins at once.

Interpretation. The model is not “small firms good, large firms bad.” It is a theory of when each form dominates once incentives to gather information are included.

3.2 Decentralization with two units of capital per division

Start with the decentralized case where each stand-alone division receives two units of capital from the external market.

If the manager becomes informed and sees one good and one bad project, he allocates both units to the good project. If he remains uninformed, each project gets one unit.

The manager’s gain from becoming informed is

$$\Delta_d^2 = \frac{g(2) - g(1)}{2}. \quad (9)$$

He chooses effort to maximize $p(e)\Delta_d^2 - \gamma e$, so the first-order condition is

$$p'(e_2^d) = \frac{\gamma}{\Delta_d^2}. \quad (10)$$

Expected total output on four units of capital is

$$y^d(4) = p(e_2^d)(g(2) + g(1)) + (1 - p(e_2^d))2g(1). \quad (11)$$

Interpretation. With two units, information is valuable because it lets the manager avoid wasting one unit on a bad project and instead concentrate both units on the good project.

3.3 Decentralization with three units of capital per division

If instead each decentralized division receives three units, the manager can avoid even worse mistakes when informed. The gain from becoming informed is

$$\Delta_d^3 = \frac{g(2) - g(1) - b(2)}{4}. \quad (12)$$

The effort choice satisfies

$$p'(e_3^d) = \frac{\gamma}{\Delta_d^3}, \quad (13)$$

and expected output on six units is

$$y^d(6) = p(e_3^d) \frac{3g(2) + g(1) + b(2)}{2} + (1 - p(e_3^d))(g(2) + g(1) + b(2)). \quad (14)$$

Since $g(2) + b(2) < 0$ and $g(1) > 0$, one gets $\Delta_d^3 > \Delta_d^2$.

Interpretation. Information is even more valuable when a manager has three units to place, because informed allocation helps avoid the especially bad outcome of putting two units into a bad project.

3.4 Lemma 1 and the decentralized credit constraint

The paper proves that if

$$3(g(2) - g(1)) + b(2) < 0, \quad (15)$$

outside investors will always prefer to give each decentralized division only two units of capital.

Interpretation. Lemma 1 fixes the decentralized benchmark at two units per division. Economically, it says that if bad two-unit investments are sufficiently painful, outsiders will not overfund decentralized firms.

3.5 Hierarchy with soft information: CEO research technology

Now suppose the two divisions are integrated into a hierarchy and information is soft. Division managers still acquire information about their own projects, but they cannot credibly pass it upward to the CEO.

The CEO can do her own research. With probability q , her research succeeds. Successful CEO research reveals whether a division is a *star*, meaning both projects are in the good state. But even when successful, the CEO cannot distinguish an average division (one good and one bad project) from a dog division (two bad projects).

Interpretation. The CEO knows less about project details than the line managers, but she may know enough to reallocate funds across divisions better than outside investors can.

3.6 Lemma 2 and the hierarchical credit constraint

Without additional assumptions, integration can relax external financing constraints because internal diversification reduces the chance that extra capital lands in a bad project. To isolate the pure

incentive trade-off, Stein imposes the sufficient condition

$$25(g(2) - g(1)) + b(2) < 0. \quad (16)$$

Under this condition, outsiders will never give the hierarchy more than four units of capital.

Interpretation. Lemma 2 deliberately shuts down the “bigger internal capital market raises more finance” channel, so the comparison focuses on incentives and reallocations, not on sheer borrowing capacity.

3.7 Hierarchy with soft information and four total units

Suppose the hierarchy receives four units of capital in total. When CEO research fails, she simply gives each division two units. When CEO research succeeds and reveals one lone-star division, she may tilt the allocation. A sufficient condition for giving all four units to the star is

$$\frac{g(2)}{2} > \frac{5g(1)}{6}. \quad (17)$$

Now look at the line manager’s incentive to become informed. The gain from successful research in a hierarchy with soft information and four total units is

$$\Delta_h^{4S} = (1 - q) \frac{g(2) - g(1)}{2} + 3q \frac{g(2) - g(1)}{8} = (1 - q) \Delta_d^2 + \frac{3q}{4} \Delta_d^2. \quad (18)$$

Hence effort satisfies

$$p'(e_4^{hS}) = \frac{\gamma}{\Delta_h^{4S}}. \quad (19)$$

Expected output is

$$y^{hS}(4) = (1 - q) \left[p(e_4^{hS}) (g(2) + g(1)) + (1 - p(e_4^{hS})) 2g(1) \right] \quad (20)$$

$$+ q \left[p(e_4^{hS}) \frac{6g(2) + g(1)}{4} + (1 - p(e_4^{hS})) \frac{3g(2) + 4g(1)}{4} \right]. \quad (21)$$

Because $\Delta_h^{4S} < \Delta_d^2$, it follows that

$$e_4^{hS} < e_2^d. \quad (22)$$

Interpretation. This is the central discouragement result. Under soft information, a line manager may lose his whole budget to another division even when his own information would have been useful. That prospect weakens ex ante research incentives.

3.8 Propositions 1 and 2: soft information can favor decentralization

Proposition 1 states that under Lemma 2 and the star-gets-everything condition,

- decentralization always yields more division-manager research effort than hierarchy:

$$e_2^d > e_4^{hS}; \quad (23)$$

- and for some parameter values decentralization also yields higher expected output:

$$y^d(4) > y^{hS}(4). \quad (24)$$

Proposition 2 weakens the allocation condition. If

$$\frac{g(2)}{2} > \frac{3g(1)}{4}, \quad (25)$$

a lone-star division still receives at least three units. Even then, decentralization continues to imply stronger line-manager incentives and may still generate higher expected output.

Interpretation. The result does not depend on the CEO literally taking everything away. Any meaningful risk of losing part or all of one's budget can weaken research incentives when information is soft.

3.9 Hierarchy with hard information

Now suppose line managers can generate hard information. If manager i exerts effort e_i , he still becomes informed with probability $p(e_i)$. But conditional on success, the information is verifiable and transmittable to the CEO with probability z .

The CEO does no separate research in this version. Instead, she processes the line managers' reports. A key assumption is that the CEO cannot observe whether a silent manager failed to learn anything or learned bad news and chose not to report it.

If a manager reports good hard information when he has it and stays quiet on bad hard information, then given a conjectured research-success probability p^c , Bayes' rule implies

$$\Pr(\{B, B\} \mid \text{quiet}) = \frac{1}{4 - 3zp^c}, \quad (26)$$

$$\Pr(\{G, G\} \mid \text{quiet}) = \frac{1 - zp^c}{4 - 3zp^c}, \quad (27)$$

$$\Pr(\{G, B\} \mid \text{quiet}) = \frac{2(1 - zp^c)}{4 - 3zp^c}. \quad (28)$$

Interpretation. Silence is not fully revealing. This gives a manager option value from research: if he uncovers good hard information he can speak up; if he uncovers bad hard information he can stay quiet.

3.10 Proposition 3: hard information strengthens research incentives

Under Lemma 2 and

$$\frac{g(2)}{2} > \frac{3g(1)}{4}, \quad (29)$$

there exists an equilibrium in which:

- managers reveal $\{G, G\}$ or $\{G, B\}$ and stay quiet on $\{B, B\}$;

- if one division reveals $\{G, G\}$ and the other is quiet, the $\{G, G\}$ division gets at least three units, and possibly four;
- expected output exceeds the decentralized benchmark:

$$y^{hH}(4) > y^d(4). \quad (30)$$

The appendix shows that the division manager's gain from being informed, denoted A^{hh} in the paper, is bounded below by

$$A^{hh} > \Delta_d^2 + \frac{z}{4}. \quad (31)$$

In fact the more detailed expression is

$$A^{hh} = \Delta_d^2 + \frac{z}{4} + \frac{z(4g(2) - 4g(1))}{16} + \frac{zp(e_j)(2g(1) - 2g(2))}{16} + \frac{z^2p(e_j)2g(1)}{16}, \quad (32)$$

which exceeds $\Delta_d^2 + z/4$ for all admissible parameter values.

Therefore line-manager effort is strictly higher in a hierarchy with hard information than under decentralization:

$$e_4^{hH} > e_2^d. \quad (33)$$

Interpretation. This is the paper's big reversal. Once information can be hardened, line managers now want to do research because good news can help them win a larger capital budget. Separation of authority from expertise now raises incentives instead of lowering them.

3.11 Proposition 4: hard information hierarchy dominates more generally

Proposition 4 relaxes Lemma 2 and allows decentralized divisions to raise two or three units and the hierarchy to raise anywhere from four to seven units, subject only to $g(2) + b(2) < 0$ and $\frac{g(2)}{2} > \frac{3g(1)}{4}$.

The conclusion is:

$$\text{if information is hard, hierarchy always strictly dominates decentralization.} \quad (34)$$

Interpretation. In the hard-information case the hierarchy wins twice:

- it reallocates better across divisions;
- and it also induces more information production within divisions.

3.12 Endogenous CEO effort and flatter organizations

Section II.B makes CEO research endogenous by writing $q = q(e_{CEO})$, where $q(\cdot)$ is increasing and concave. In the soft-information case, with four total units and the star-gets-everything condition, the CEO's gain from successful research is

$$\Delta_{CEO} = p(e_4^{hS}) \frac{2g(2) - 3g(1)}{4} + (1 - p(e_4^{hS})) \frac{3g(2) - 4g(1)}{4}. \quad (35)$$

Differentiating yields

$$\frac{d\Delta_{CEO}}{dp(e_4^{hS})} = \frac{g(1) - g(2)}{4} < 0. \quad (36)$$

Interpretation. CEO effort is less valuable when division managers work harder. Thus CEO effort and division-manager effort are strategic substitutes under soft information. This opens the door to flatter organizations and even multiple equilibria, such as a control-freak equilibrium versus a laissez-faire equilibrium.

3.13 Endogenous hardness and excess bureaucracy

Section II.C makes hardness endogenous. A line manager can either pursue soft information or hard information. If he spends one unit of effort on soft information, success arrives with probability p_s . If he spends one unit of effort on hard information, success arrives with probability

$$p_h = \beta p_s, \quad \beta < 1, \quad (37)$$

and conditional on success the information is verifiable with probability z .

Interpretation. Hard information is less productive to generate, but privately more attractive inside a hierarchy because it can influence the manager's capital budget. This is why hierarchy may create too much documentation and too much internal report writing relative to the social optimum.

4 Identification and comparison logic (what makes the results credible)

4.1 What exactly is being compared

The paper is not empirically identifying a causal effect from data. Instead, it analytically compares equilibrium outcomes under different organizational regimes using the same primitive environment.

Interpretation. The analogue of “identification” here is the clean isolation of mechanisms. Stein asks: if we hold the technology and the financing frictions fixed, how does changing organizational design alter information production and capital allocation?

4.2 What assumptions drive the soft-information result

The decentralization result under soft information needs four ingredients:

- outside investors cannot directly elicit truthful private information;
- the CEO can sometimes reallocate capital across divisions better than the outside market;
- line managers cannot credibly pass their own project information upward;
- and the CEO cannot commit ex ante to not using her authority once she becomes informed.

Interpretation. If any of these assumptions were removed, the anti-hierarchy result under soft information would weaken or disappear.

4.3 Why the CEO can beat the outside market

The paper is explicit that the CEO's informational edge over outside financiers is an assumption that needs justification. Stein offers two justifications:

- dispersed outsiders face free-rider problems in information acquisition;
- even a single outsider may do less research because authority and information acquisition are complements.

Interpretation. A CEO with strong control rights has more to gain from becoming informed, because she can not only lend or withhold capital, but also intervene in divisional operations more broadly.

4.4 Why lack of commitment matters

A central idea is that the CEO cannot credibly promise:

- to remain uninformed,
- or, once informed, to refrain from reallocating capital.

Interpretation. This is what makes hierarchy costly under soft information. If the CEO could commit to leaving each division with at least two units, line managers would not fear losing the chance to use their expertise.

4.5 Why hard information reverses the result

The hard-information case reverses the sign because the line manager now gets an extra source of private benefit from research:

- under decentralization, research only improves the return on a fixed budget;
- under hierarchy with hard information, research can also help the manager obtain a larger budget.

Interpretation. Once this gross-budget effect becomes active, hierarchy does not merely preserve incentives; it can amplify them.

4.6 Why Lemma 2 is useful

Lemma 2 is not meant to say that integrated firms usually cannot raise more capital. Quite the opposite: Stein notes that hierarchy often can relax financing constraints. Lemma 2 is imposed only to strip that channel away temporarily.

Interpretation. This is a methodological device. It prevents the reader from thinking the soft-information result is driven by a hidden change in total available capital rather than by incentive effects.

4.7 Why CEO effort and line-manager effort can conflict

In the soft-information extension with endogenous CEO effort, the CEO sometimes over-tilts capital toward a star division because her information is coarse. The cost of this activism is bigger when division managers are already informed and can use capital well inside their divisions.

Interpretation. This is why the efforts of senior and junior managers become strategic substitutes. Hierarchy can fail not only because bosses discourage subordinates, but also because capable subordinates reduce the marginal value of boss intervention.

4.8 What the model can and cannot distinguish

The model is strongest on comparative statics:

- softer information makes decentralization and flatter structures more attractive;
- harder information makes hierarchy more attractive.

But the model does *not* fully distinguish among several deeper interpretations:

- whether the CEO's advantage comes mainly from authority, free-rider problems, or both;
- whether small-business lending is really about softness of information or about some omitted contractual friction;
- whether all forms of hard information are equally easy for CEOs to process.

Interpretation. The paper is a mechanism paper, not a full descriptive theory of all large organizations.

4.9 Relation to the literature

The paper sits between several literatures:

- It extends Stein (1997) by endogenizing line-manager research incentives.
- It is related to Aghion and Tirole's concern that formal authority can discourage subordinate initiative.
- It connects to the influence-cost literature, but differs sharply because advocacy based on genuinely hard information can be productive rather than wasteful.
- It also differs from pure communication-network theories by placing authority and capital allocation at the center of the problem.

Interpretation. The novel contribution is not just "bosses discourage workers." It is that whether this happens depends on the hardness of the information environment.

5 Main propositions and empirical implications

5.1 Lemma 1: decentralized firms remain externally constrained

Read:

- If $3(g(2) - g(1)) + b(2) < 0$, outside investors always choose two rather than three units for each decentralized division.
- So the decentralized benchmark becomes two units per division, four in total.

Economic message. This keeps the decentralized case from being overfunded when uninformed managers might accidentally place too much capital into bad projects.

5.2 Lemma 2: the hierarchy is held to the same four-unit benchmark

Read:

- If $25(g(2) - g(1)) + b(2) < 0$, even an integrated hierarchy never receives more than four units of capital from outsiders.
- This temporarily removes hierarchy's usual external-finance advantage.

Economic message. The soft-information comparison is intentionally conservative for hierarchy. If hierarchy still loses under these assumptions, the loss really comes from distorted incentives.

5.3 Propositions 1 and 2: soft information can make decentralization superior

Read:

- Under soft information, line managers in a hierarchy work less hard because the CEO may reallocate capital away from them.
- Decentralization always gives stronger line-manager incentives:

$$e_2^d > e_4^{hS}.$$

- For some parameter values, decentralization also produces higher expected output.
- This remains true even if the CEO only partially tilts the budget toward a star division rather than moving all four units.

Economic message. Under soft information, hierarchy buys better across-division reallocations at the price of weaker within-division information production. When the latter cost is large enough, small autonomous units are better.

5.4 Proposition 3: hard information reverses the sign

Read:

- When division managers can generate verifiable hard information, they reveal good news and suppress bad hard news.

- The CEO reallocates toward divisions that can document strong prospects.
- A division manager's gain from being informed becomes larger than under decentralization by at least $z/4$.
- Hence line-manager effort is stronger in the hierarchy, and total expected output is also higher.

Economic message. Hard information turns division managers into advocates for their units. Now hierarchy improves both internal reallocations and information production.

5.5 Proposition 4: with hard information, hierarchy dominates more generally

Read:

- Even when one relaxes Lemma 2 and allows for richer external-financing possibilities, hierarchy still dominates decentralization in the hard-information case.
- This holds whether decentralized divisions can raise two or three units and whether the hierarchy can raise four to seven units.

Economic message. The hard-information result is not a knife-edge artifact of the four-unit benchmark. It is a robust consequence of verifiable advocacy inside the firm.

5.6 Section II insights: dis-integration, flatter firms, and bureaucracy

Read:

- If there are scope synergies X across divisions, full breakup may be too costly even when decentralization is attractive from an incentive perspective.
- In that case firms may stay integrated but try to mimic decentralization using internal devices that make meddling harder.
- When CEO effort is endogenous, soft information favors flatter organizations because senior and junior efforts are strategic substitutes.
- When hardness is endogenous, hierarchies may induce too much documentation and too many reports.

Economic message. The model does more than compare “two small firms” to “one big firm.” It also explains why real organizations may use flatter structures, holding-company forms, or administrative barriers to internal capital transfers.

5.7 Section III: the banking evidence

Read:

- Small banks devote a larger share of assets to small-business lending than large banks.

- When large banks acquire small banks, small-business lending tends to fall sharply.
- The loans that disappear after mergers do not appear to be simply bad loans: many are picked up by other local banks, and loan severance is not strongly tied to borrower credit quality.
- The discouraging effect seems linked not just to size, but to organizational complexity: more branches and multibank holding-company status are associated with less small-business lending.
- Large banks appear more likely to rely on standardized financial-statement criteria, while small banks lean more on character-based assessments.

Economic message. These patterns line up closely with the model's interpretation of relationship lending as an activity that depends heavily on soft information produced far from the ultimate decisionmaker.

6 Short summary

- The paper studies how organizational form affects the production of information and the allocation of capital.
- Under decentralization, line managers keep authority over their own capital budgets, which strengthens incentives to produce soft information.
- Under hierarchy, a CEO can reallocate capital across divisions more intelligently than outside investors, but this may discourage line-manager research if their information cannot be credibly transmitted.
- Hence with soft information, decentralization may dominate hierarchy.
- If information can be hardened and passed upward, the result flips: hierarchy both reallocates capital better and induces more line-manager research.
- The paper also predicts that soft information favors flatter organizations and that hierarchy may generate excess bureaucracy when documentation is privately rewarded.
- The motivating application is bank consolidation and the decline of relationship-based small-business lending after mergers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the relative efficiency of decentralization and hierarchy depends critically on the communication properties of information. A hierarchy is not automatically superior just because it can move resources across projects, and decentralization is not automatically superior just because it gives more local autonomy. The central question is whether the information produced by line managers can travel upward inside the organization.

7.2 Economic intuition

The cleanest way to read the paper is as a theory of the separation of authority and expertise.

- When information is soft, separating authority from expertise is costly because the expert may lose the ability to act on what he knows.
- When information is hard, separating authority from expertise is useful because the expert can persuade the authority holder and thereby obtain more resources.

So the same hierarchy that discourages initiative in one environment can encourage it in another.

7.3 Takeaway

The deepest message of the paper is that organizational diseconomies are information-contingent. Big hierarchical firms are most likely to struggle in activities where valuable information is local, qualitative, and hard to verify. They are most likely to excel when information can be documented, transmitted, and aggregated at the top. That is why the paper remains such a useful framework for thinking about banks, conglomerates, and any organization that must decide where expertise should sit relative to control over capital.